



- ◉ Family Leguminosae is divided into three sub families
- ◉ 1. Caesalpinaceae
- ◉ 2. Fabaceae
- ◉ 3. Mimosiaceae

COMMON PLANT

- ◉ *Pisum sativum*
- ◉ *Cicer arietinum*
- ◉ *Glycine max*
- ◉ *Lathyrus odoratus*
- ◉ *Crotolaria gunceria*
- ◉ *Arachis hypogaea*

DISTRIBUTION

- ◉ The family is also known as Fabaceae. It includes 600 genera and 1200 species.
- ◉ It is regarded as the second largest family of dicotyledons.
- ◉ In Nepal the family is represented by 94 genera and 379 species.

HABIT

- ◉ Fabaceae a wide variety of habits.
- ◉ The plants may be herbs eg *Trifolium*
- ◉ Climbers eg *Pisum sativum*
- ◉ trees *Erythrina* aquatic plant.



- ◉ The family is economically very important because pulses are obtained from many members.
- ◉ Habitat:
- ◉ Mostly they are mesophytic. A few may be xerophytic and hydrophytic.
- ◉ Vegetative characters:
- ◉ Root: Tap root branched with modified root as root nodules to have nitrogen fixing bacteria e.g. *Rhizobium* sp.

- ◉ Stem: herbaceous, weak or woody branched cylindrical or angular, solid or fistular, hairy or glabrous.
- ◉ Leaves: Cauline and ramal, Alternate, rarely opposite or whorled, pinnately compound, stipulate, petiolate, a leafbase is swollen (pulvinus), upper leaflet is modified into tendril.

INFLORESCENCE

- ◉ Raceme or panicle raceme, sometimes solitary axillary.
- ◉ **Flowers**
- ◉ Flowers are bracteates, bracteolate, pedicillate, complete, bisexual, zygomorphic usually Hypogynous or slightly perigynous, pentamerous, papilionaceous.
- ◉ **Calyx**
- ◉ Usually 5 sepals, gamosepalous, aestivation is valvate or imbricate, odd sepal is anterior, usually persistent, sepaloid, inferior.

- ◉ **Corolla:**
- ◉ 5 petals, polypetalous, corolla is papilionaceous.
- ◉ The odd posterior petal is called standard or Vexillum is the largest petal.
- ◉ The two lateral ones, called wing, two innermost are smallest united to shaped structure called Keel.
- ◉ Descending imbricate, inferior.

ANDROECIUM

- ◉ Stamens 10, usually diadelphous e.g. 1+ (9), sometimes monadelphous, rarely polyandrous, anthers are dithecal, dorsifixed and introrse.
- ◉ **Gynoecium:**
- ◉ Monocarpellary, ovary superior, unilocular with many ovules at ventral sutures.
- ◉ Placentation is marginal. The style is long stigma capitate.
- ◉ **Fruit:**
- ◉ It is a legume (pod) or lomental.
- ◉ Cross pollination by insect.
- ◉ Seed : Ex-albuminous

◉ Diagnostic features

1. Tap root, modified into root nodules with N₂ fixing bacteria.

2. Pinnately compound leaves with leaf tendrils.

3. Inflorescence Raceme

4. Flower pentamerous

5. Corolla Papilionaceous

Stamen Diadelphous, grouped in two bundles, 9 in one and one is single.

Unilocular ovary, marginal placentation.

6. Fruit : legume or Pod.

FLORAL DIAGRAM

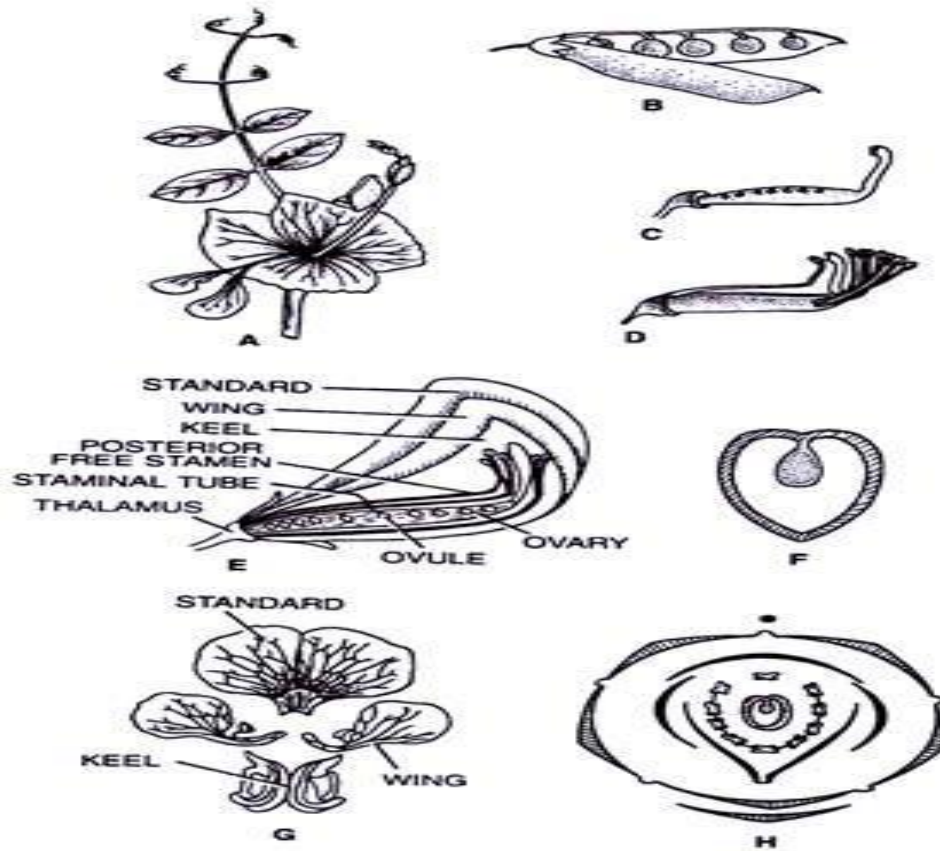


Fig. 52.1. *Lathyrus odoratus* – A. twig with flowers; B. dehiscent pod; C. ovary with style and stigma; D. diadelphous stamens with staminal tube covering the ovary; E. V.S. flower; F. T.S. ovary; G. corolla; H. Floral diagram.

Floral formula

Br, op $\text{\text{K}(5)}$ C1+2+(2) A(9)+1 G_1

ECONOMIC IMPORTANCE

- ◉ **Plants used as food:**
- ◉ the family contains many food plants including *Cajanuscajans* *Pisum sativum* ,*Glycine max* ect.
- ◉ Plant used for oils *Arachis hypogea* *Glycine max*.
- ◉ Plant used ornamental purposes *Sesbania grandiflora* *Lathyrus odoratus* *monosperma* etc.
- ◉ plants timber and fibres Shishum (*Dalbargia sisso*), Sunn hemp (*Crotolaria juncea*).



Thank
You.....